

Another way to think about Gateway API

Thibaut Lapierre

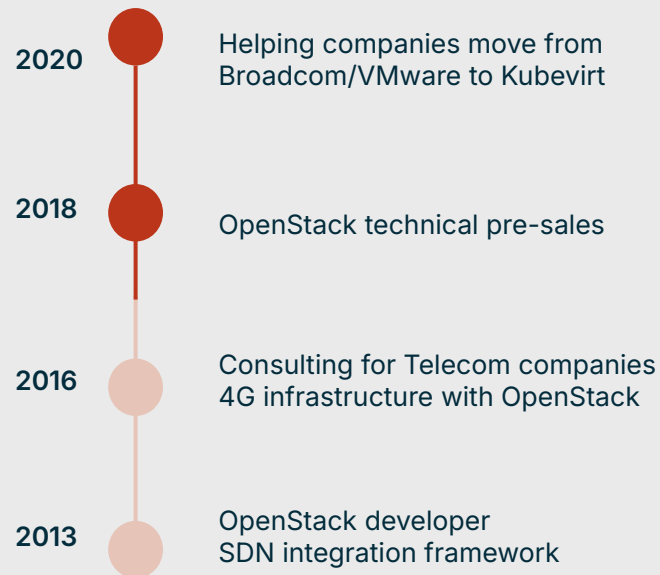
Who's speaking ? (unimportant but relevant for what comes next)



Thibaut Lapierre



<https://blog.epheo.eu> | <https://github.com/epheo>



[pkg/netutils/subnet_allocator.go](https://pkg.netutils/subnet_allocator.go) 19-81

Why is k8s networking model broken for IaaS

The Pod Network

A single Network for all your workload and namespaces

One subnet per worker node
-> L3 routed IPAM

No self service

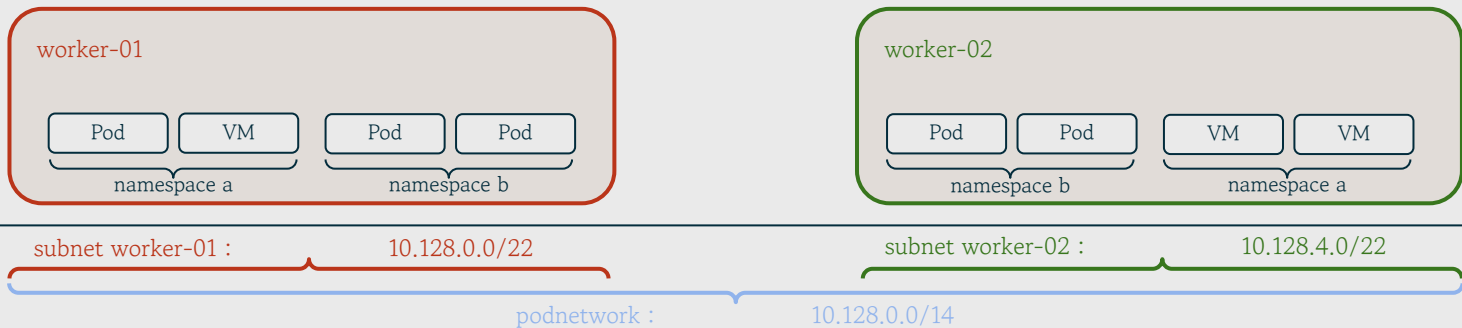
Network Policies

Isolation by micro segmentation

Complex to handle and maintain
(prone to mistakes)

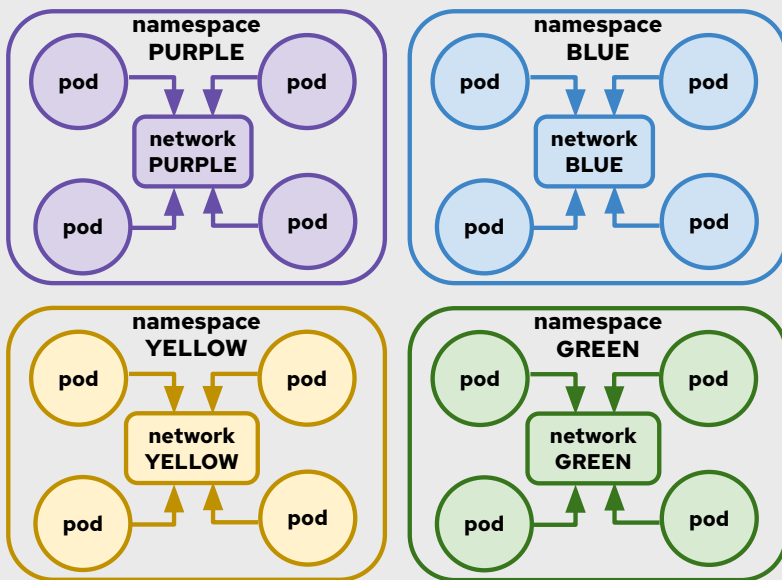
You have to trust they are well written

Infrastructure as a Service aim to replicate the baremetal world and consumers are used to the VPC model.





What does exist today



I guess it all comes from laziness

OVN User Defined Networks

The ability to create private networks.

Can be ns scoped or span multiple namespaces

Can replicate pod network subnet split or span L2

Can be routed or fully private

But yet...

We lose the ability to expose resources between UDN

IaaS use cases usually requires routes and proxy between those private networks

OKEP-5224



What we should improve

OKEP-5224: Connecting UserDefinedNetworks

Pushed by Nvidia, developed by Red Hat

Pure L3 connectivity between routed UDN

Do not address secondary (private) UDN

Two tenants' Primary UDNs cannot talk east-west



Need virtual routers



Primary UDN only



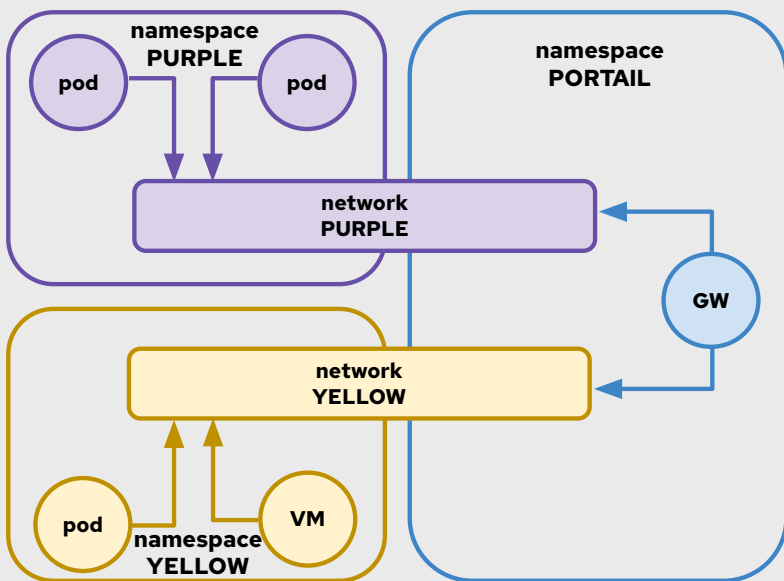
Need to address L4-L7 between UDN

C'est bien, mais c'est pas assez...

Portail



Another way to think about Gateway API



Portail



Wanted to learn about io_uring and eBPF (didn't go well..) Re-wrote with tokio-rs



Customers asking for a way to expose services from isolated namespaces without going back to ToR switch or BGP



Waiting for OKEP-5224



Use the Gateway API to proxy services north/south and east/west equally

<https://github.com/epheo/portail>

Obstacles and current issues

Performances	Proxying is not routing, and while it address additional layers, performances are impacted	1	K8s networking performances are poor anyway
		2	600k RPS for http with portail
Secondary UDN only	One pod can't bridge two tenants' Primary UDNs. Three platform guards forbid attaching a Primary as a secondary interface (Multus namespace isolation, OVN-K role guard, degenerate NAD aliasing)	1	Maybe that's enough: OKEP-5224 for Primary, Gateway API for e/w secondary net.
		2	Finding a better way for primary UDN
TCP / UDP are experimental API	OpenShift currently only ship (and owns) Gateway API v1.4.1 standard. CRDs are owned and reconciled by the OpenShift Ingress Operator	1	Do not impact vanilla k8s anyway
		2	This will naturally change in future release of OpenShift

demo-effect

→ A broken demo is
always better than a
long speech

1

Test Primary
UDN isolation

2

Create a
Gateway
(with Portail)

3

Create E/W
listeners

4

Test traffic
via GW